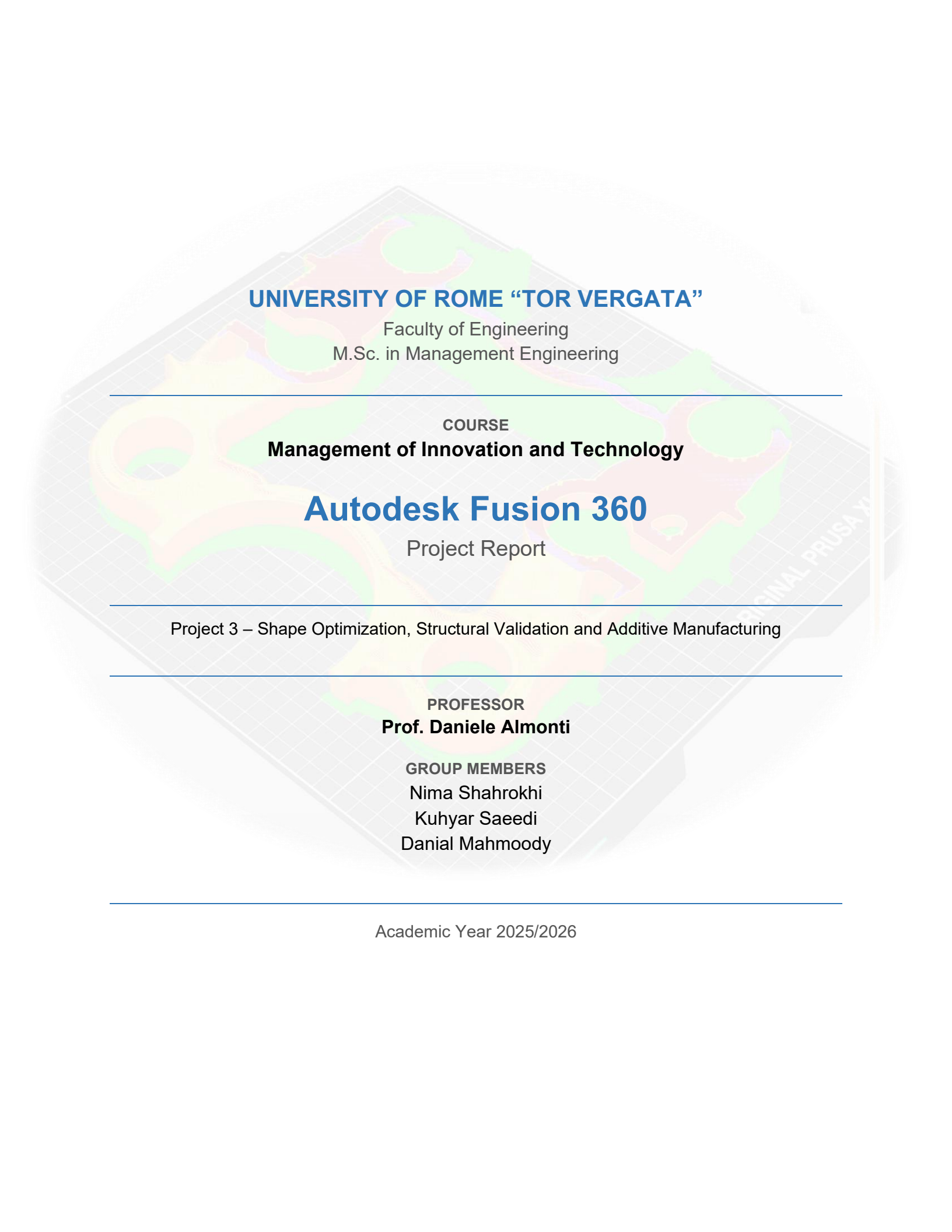




UNIVERSITY OF ROME “TOR VERGATA”

Faculty of Engineering
M.Sc. in Management Engineering

COURSE

Management of Innovation and Technology

Autodesk Fusion 360

Project Report

Project 3 – Shape Optimization, Structural Validation and Additive Manufacturing

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1. Introduction and Objective

The aim of this project is to redesign an existing steel component so that it becomes as lightweight as possible while remaining structurally adequate for its working loads. Rather than removing material by hand, the geometry is reworked using a shape (topology) optimization study, which automatically removes material from the regions that contribute little to the stiffness of the part, leaving an efficient load-carrying structure behind.

The complete workflow followed in this project is made up of four consecutive phases:

- Shape optimization – the original solid is lightened down to a target mass while its stiffness is maximized under the applied loads and boundary conditions.
- Geometry reconstruction – the optimized result, which is delivered as a mesh, is repaired and converted back into a usable solid (organic) body.
- Static structural validation – a separate static stress study confirms that the lightened part can still carry the design loads with an acceptable margin.
- Additive manufacturing preparation – the validated geometry is set up for 3D printing and finally checked in a dedicated slicer to evaluate its real manufacturability.

An important point of the project is the link between the optimized shape and the manufacturing method. The organic, free-form geometry produced by the optimizer cannot be obtained sensibly with conventional processes (milling, turning, casting); it is essentially a shape that only additive manufacturing can realize. The optimization is therefore not an academic exercise alone – it is a genuine design-for-additive-manufacturing exercise, and this is the reason the whole study is built around a 3D-printed final part.

2. Starting Geometry

The starting point is the solid component that was modelled in Fusion from the technical drawing provided by the professor. It is a compact, approximately square steel plate carrying four through bores at its corners (the functional holes of the part) and two semicircular notches on opposite edges. This solid body is the input of the optimization: it represents the maximum allowable design space, i.e. the volume inside which the optimizer is free to keep or remove material.

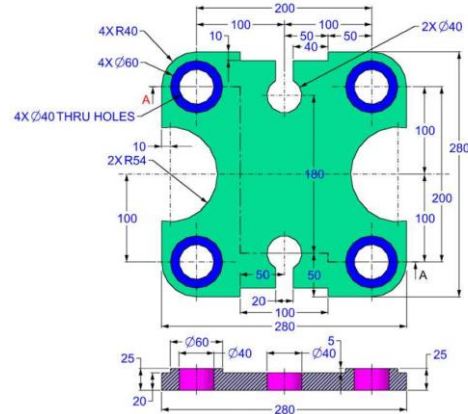


Figure 1. Original solid component (target body) used as the design space for the optimization.

Because the optimizer needs a single, well-defined body to work on, this component is selected as the Target Body of the study. The four bores and the contact areas around them are the features that must survive the optimization, since they define how the part is mounted and loaded; this requirement is enforced later through the preserved regions (Section 3.5).

3. Shape Optimization Study – Setup

The optimization is carried out in the Simulation workspace of Autodesk Fusion. A new study of type Shape Optimization is created; Fusion describes this study as a tool to “optimize parts to make them lightweight and structurally efficient based on the loads and boundary conditions applied to the geometry”, which is exactly the intent here. The study uses the Metric (SI) unit system throughout.

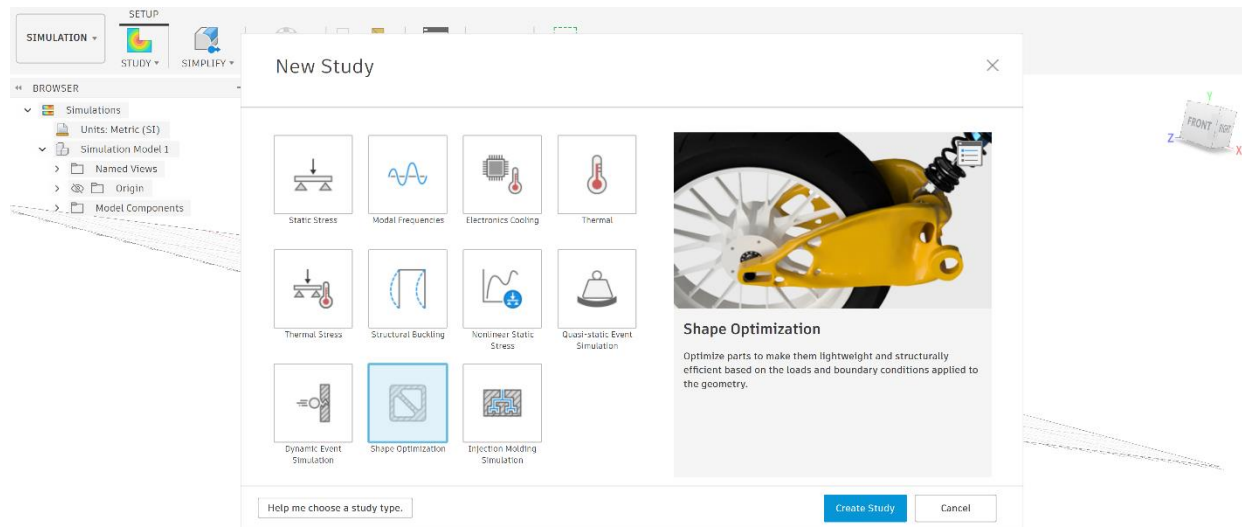


Figure 2. Creation of the Shape Optimization study in the New Study dialog.

3.1 Material

The component is assigned a steel material (category: Metal). In a shape optimization study the safety factor is evaluated with respect to the yield strength of the material: the optimizer keeps the part below the yield limit divided by the chosen safety factor, so the surviving structure is dimensioned against permanent (plastic) deformation rather than against final rupture.

Component	Material / category	Safety factor reference
Simulation Model 1	Steel – Metal	Yield Strength

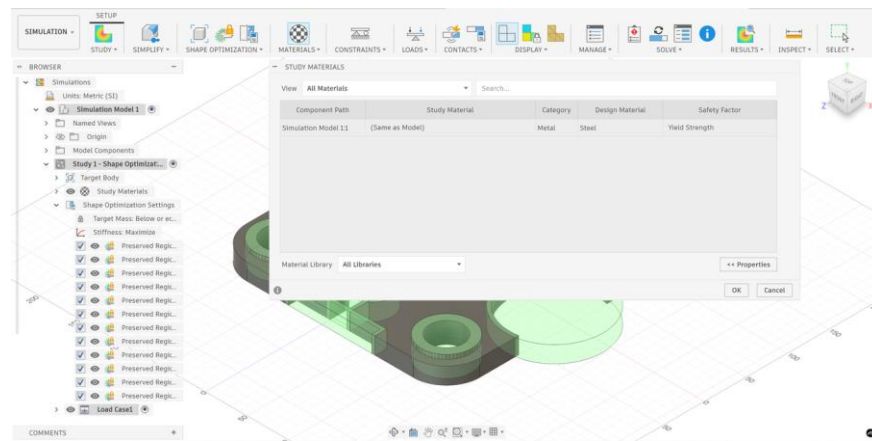


Figure 3. Study material: steel, with the safety factor referenced to yield strength.

3.2 Target Body

The original solid plate is selected as the single Target Body of the study. This is the body on which material will be added or removed; everything outside this body is ignored by the optimizer.

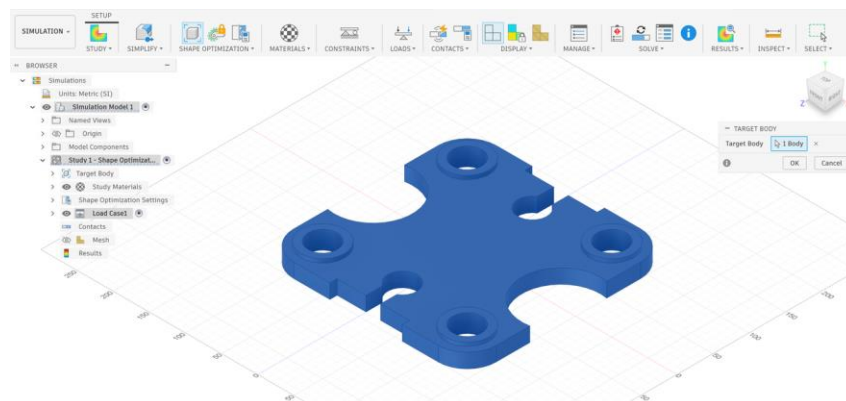


Figure 4. Selection of the target body (1 body).

3.3 Optimization Objectives and Criteria

The behaviour of the optimizer is defined through the Shape Optimization Criteria dialog, split into global objectives and global constraints:

Parameter	Expression	Value
Target Mass (objective)	$\leq$ (below or equal)	50.00 %
Stiffness (objective)	Maximize	—
Preserve Boundary 1–3	N/A	—
Preserve entities with loads / constraints	Enabled	—

Two objectives are therefore active at the same time. The Target Mass objective instructs the optimizer to keep only up to 50% of the original mass – i.e. a target weight reduction of about 50% – while the Stiffness objective is set to Maximize, so that the remaining material is arranged to make the part as rigid as possible under the load case. The option “Preserve Entities with Loads and Constraints” is enabled, which guarantees that the faces where forces and supports are applied are never deleted, keeping the boundary conditions physically meaningful.

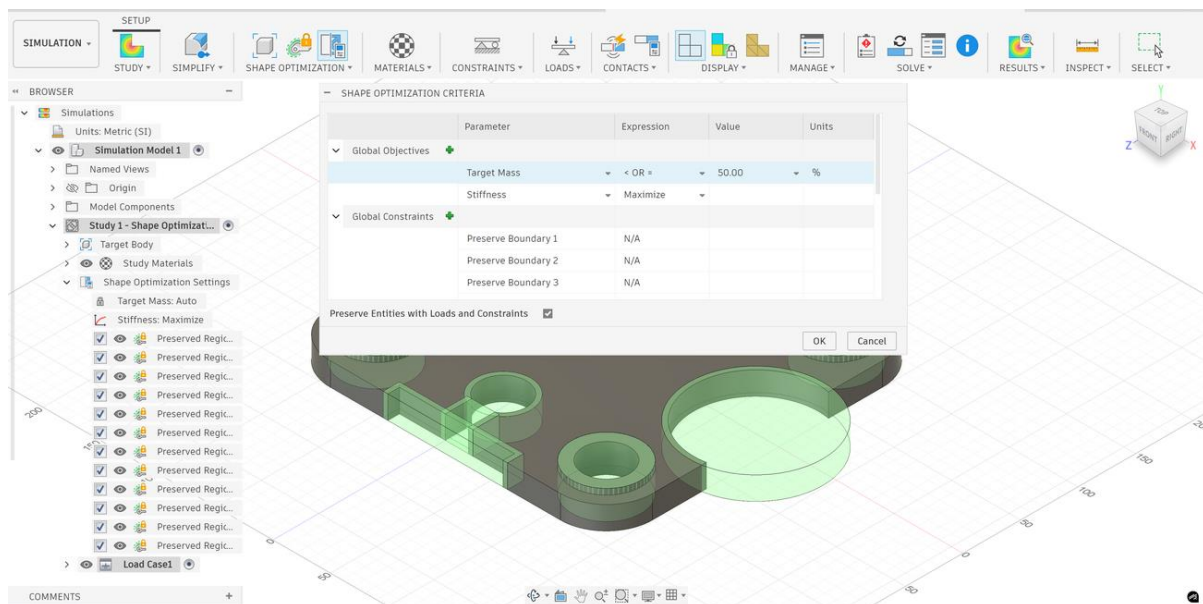


Figure 5. Optimization criteria: target mass $\leq 50\%$ with stiffness maximized.

3.4 Preserved Regions

In addition to the automatic preservation of loaded and constrained faces, a set of preserved regions is defined manually so that the optimizer cannot remove material from the functional zones of the part. Each preserved region is created from a face and given a cylindrical boundary aligned with the corresponding bore, which protects the hole and the material immediately around it. Several such cylinders are placed over the four bores and their surrounding pads, as summarized below.

Boundary shape	Shape direction	Boundary size (example values)
Cylinder	Aligned	Radius 30 mm, axial extent 25 mm
Cylinder	Aligned	Radius 60 mm, axial extent 20 mm

The green volumes visible around the holes represent these protected regions: material inside them is locked and survives the optimization regardless of how aggressively the rest of the body is hollowed out. This is what keeps the bores circular and usable in the final part.

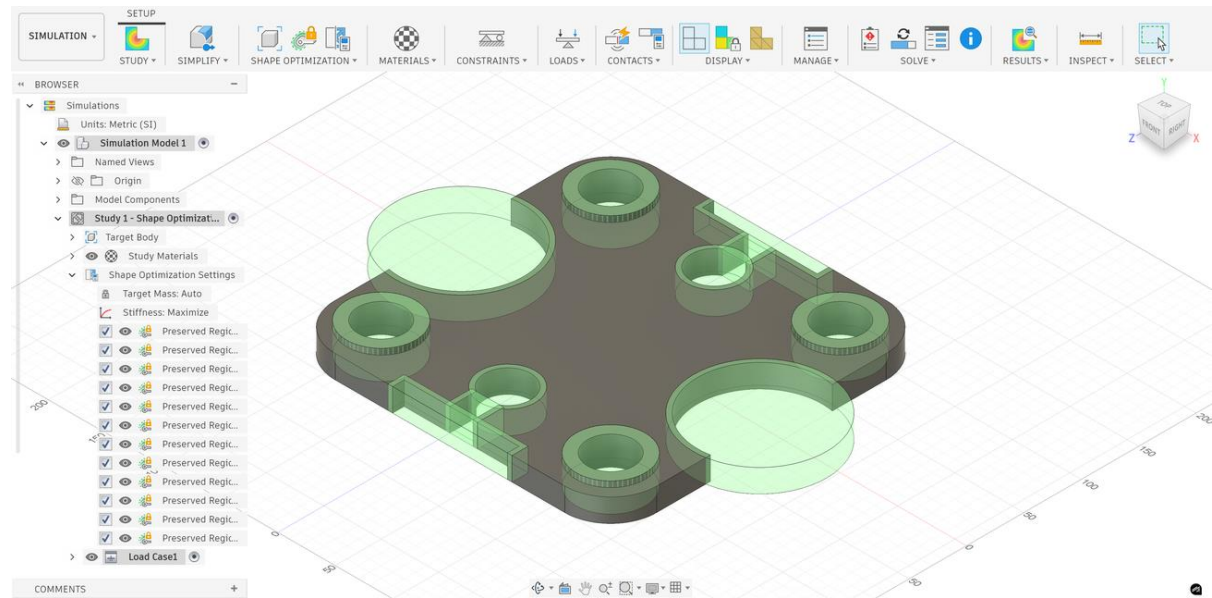


Figure 6. Preserved cylindrical regions (green) defined over the functional bores.

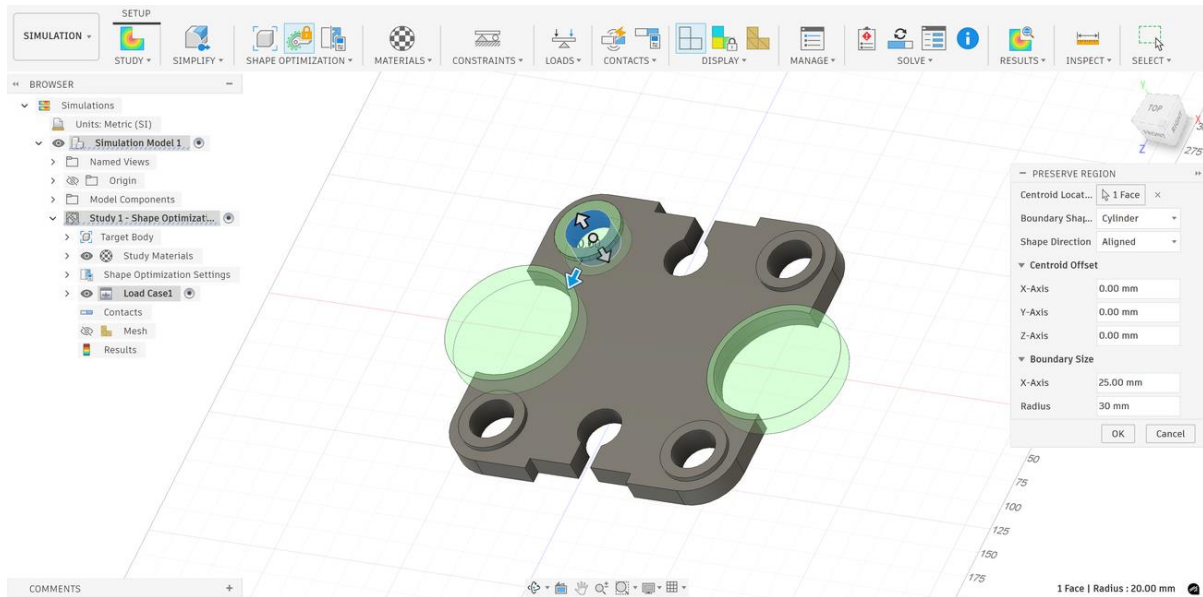


Figure 7. Preserve Region dialog showing a cylindrical, aligned boundary.

3.5 Structural Constraints

To anchor the part in space, a Fixed constraint is applied to four faces – the inner faces of the bores. The constraint locks all three translational degrees of freedom (U_x , U_y and U_z), so these faces are held completely fixed and act as the reaction points of the structure.

Constraint type	Target	Locked axes
Fixed	4 faces (bore faces)	U_x , U_y , U_z

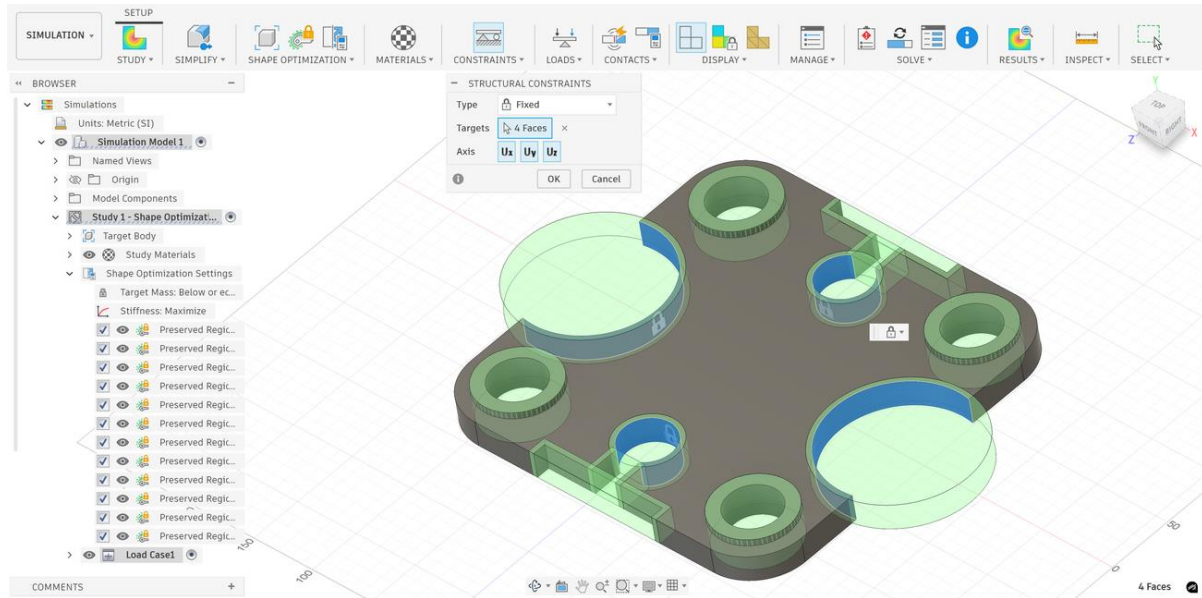


Figure 8. Fixed constraint applied to the four bore faces (all DOF locked).

3.6 Structural Loads

The working load is represented by four forces, one applied to each bore face, grouped under a single load case (Load Case1). Each force has a magnitude of 1000 N – a value of the order of a 100 kg load, chosen to be large enough to produce a meaningful structural response without being unrealistic for a part of this size. The forces act in the plane of the plate, oriented along the diagonal directions (defined by X-angles of 45° and 135°), so that the loads pull the bores towards / away from the centre and exercise the connecting structure between them.

Load	Type	Target	Magnitude / direction
Force1–Force4	Force	Bore faces (≈ 20 mm radius)	1000 N each, in-plane ($45^\circ / 135^\circ$)
Gravity	Body load	Whole body	Disabled

Gravity is deliberately switched off. For a small component the self-weight is negligible compared with the applied forces, so including it would only add noise to the analysis; this is consistent with the recommended approach of neglecting gravity at this scale and relying on the externally applied loads to drive the optimization.

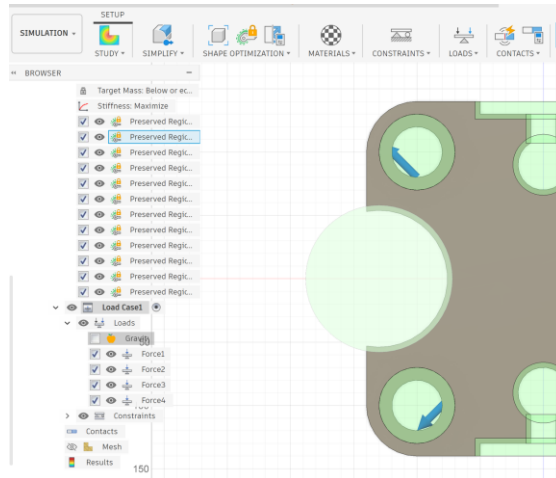


Figure 9. Load case: four 1000 N in-plane forces on the bore faces (gravity disabled).

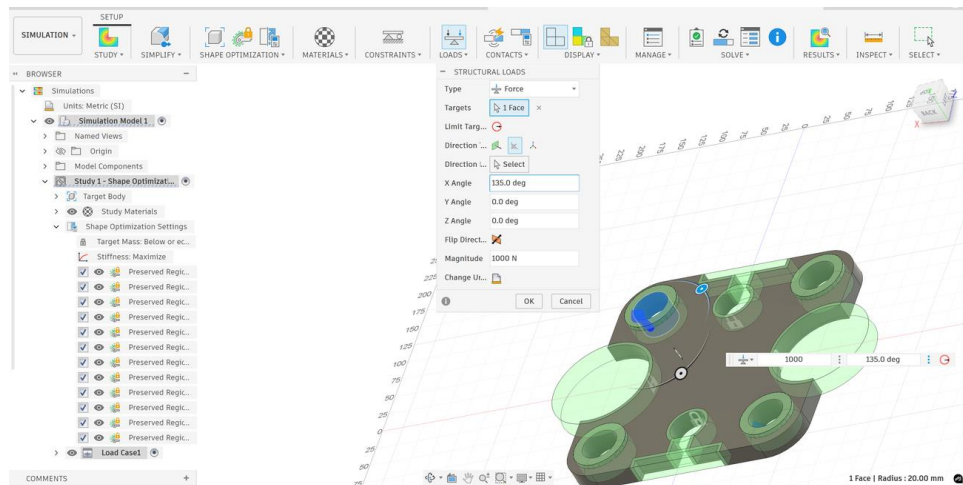


Figure 10. Structural load dialog (Force, 1000 N, direction defined by angle).

3.7 Degrees-of-Freedom Check

Before solving, the model is verified with the Degrees of Freedom display, which colours the body according to how well it is restrained: green = fully fixed, cyan = potentially fixed, yellow = partially fixed, red = free. The whole part appears green, meaning it is fully constrained with no rigid-body motion left. This confirms that the constraints are sufficient and the problem is well posed, so the solver will not fail on an under-constrained model.

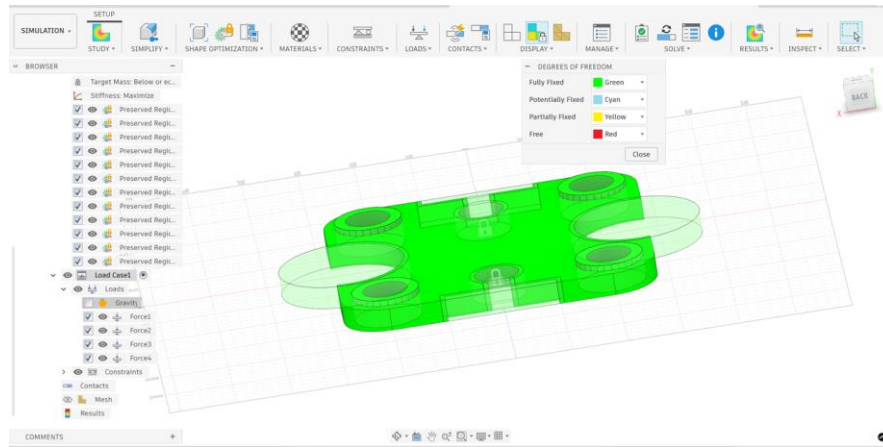


Figure 11. Degrees-of-freedom check: the part is fully fixed (entirely green).

3.8 Mesh

A finite-element mesh is generated for the optimization. The mesh uses a model-based average element size set near the fine end of the range (of the order of 1% of the model size). A finer mesh discretizes the design space more accurately, which lets the optimizer resolve the load paths in more detail and produces a smoother, better-defined optimized shape, at the cost of a longer solve. When the mesh has not yet been computed Fusion prompts to build it, and it is generated before launching the solution.

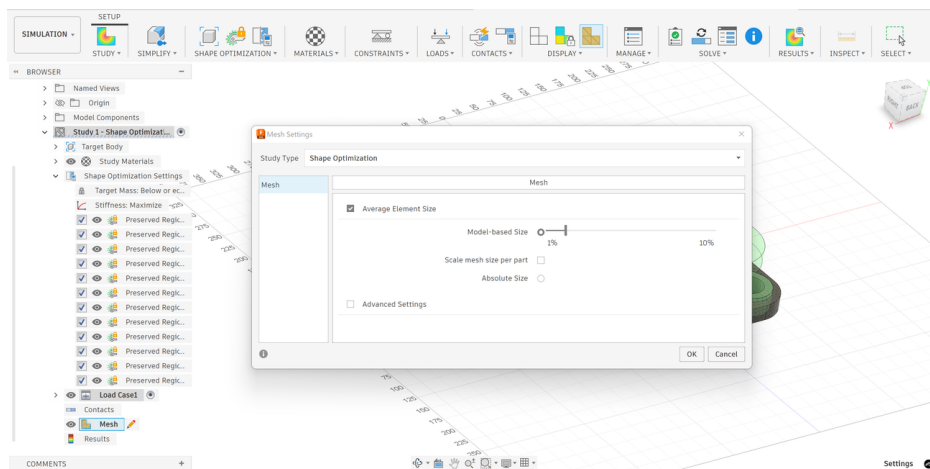


Figure 12. Mesh settings (model-based average element size).

3.9 Solve

With material, target body, objectives, preserved regions, constraints, loads and mesh all defined and the model reported as Ready, the study is launched. The solution runs on the cloud and consumes 3 cloud credits. The optimizer iteratively removes material from the low-stress regions while respecting the 50% target mass and the preserved zones, and returns the optimized (mesh) geometry that is taken forward to the reconstruction phase.

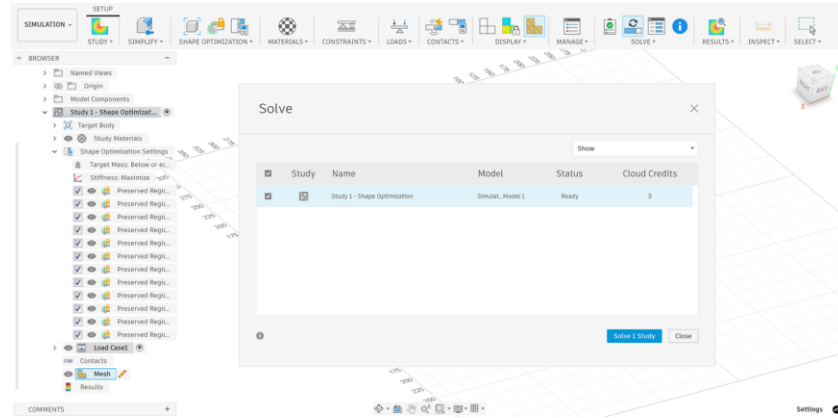


Figure 13. Solve dialog: study ready, 3 cloud credits.

4. Optimization Result

Once the study is solved, Fusion returns the optimized geometry together with a colour map that describes how efficiently each region of material is being used. The legend runs from 0 to 1 with a Target threshold: material in the warm colours (towards red, close to 1.0) is strongly involved in carrying the load and must be kept, while material that falls below the target line contributes little and is removed. The translucent light-green volumes around the bores are the preserved regions defined during setup, which remain intact. The result is the expected organic skeleton: four solid hubs around the bores connected by curved, branch-like ribs that follow the diagonal load paths, with the central low-stress area emptied out.

The optimization reports the achieved lightweighting directly:

Quantity	Value	Note
Target mass (objective)	$\leq 50 \%$	Imposed in setup
Mass ratio (result)	51.22 %	Fraction of the original mass retained

Quantity	Value	Note
Approximate mass	4.582 kg	Mass of the optimized body
Estimated mass reduction	≈ 49 %	1 – mass ratio

The retained mass (51.22%) is essentially on the 50% target, so the part has been made roughly half as heavy while keeping the stiffness the optimizer was asked to maximize. Taking the retained fraction back to the original gives a starting mass of about 8.95 kg, confirming a weight saving of close to one half. The optimized result is then promoted from the simulation into the Design workspace as a mesh object, so it can be reworked into a usable model.

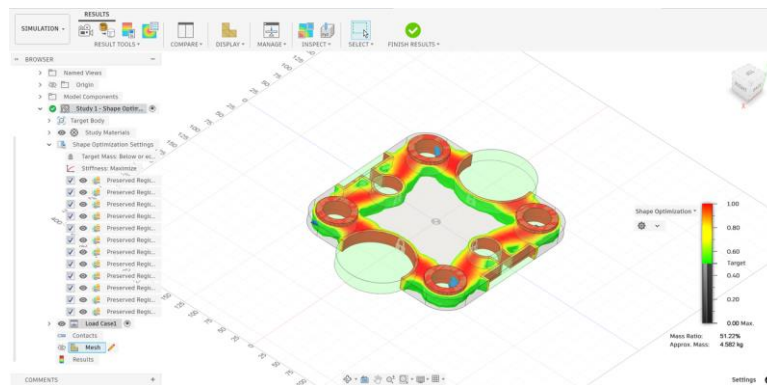


Figure 14. Optimization result with the colour map; Mass Ratio 51.22%, Approx. Mass 4.582 kg.

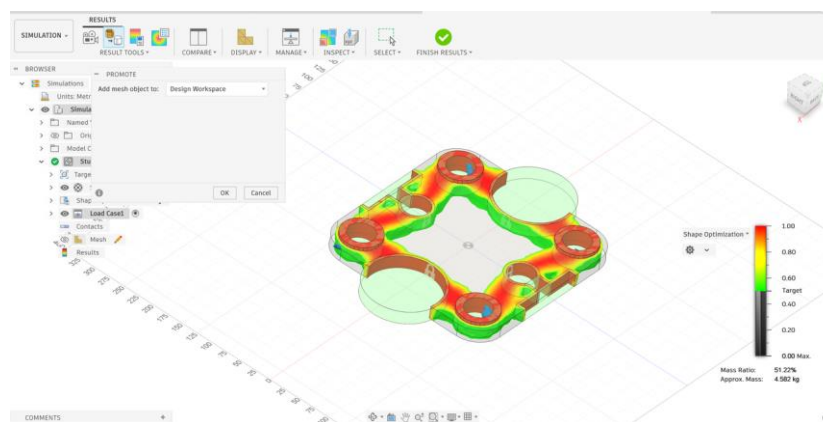


Figure 15 – Promoting the optimized mesh into the Design workspace.

5. Geometry Reconstruction

The promoted geometry is a mesh (a triangulated surface), not a normal solid body, and at this stage it is coarse and faceted. A mesh of this kind cannot be used directly for further engineering work, so it has to be repaired and then converted into a proper solid body before it can be validated and manufactured. The reconstruction is carried out in the Mesh and Solid environments of the Design workspace.

5.1 Mesh Repair

The first step is to repair the mesh. Because the optimized surface is built from a large number of triangles, some facets may be inconsistently oriented or the shell may contain small defects, which would make a clean conversion impossible. The Repair command is run with the Rebuild repair type (Adaptive, density 128) and Quick Analysis enabled, which rebuilds the triangulation into a single, correctly oriented and watertight shell. After this step the mesh reports a single shell and a single face group, confirming it is closed and consistent.

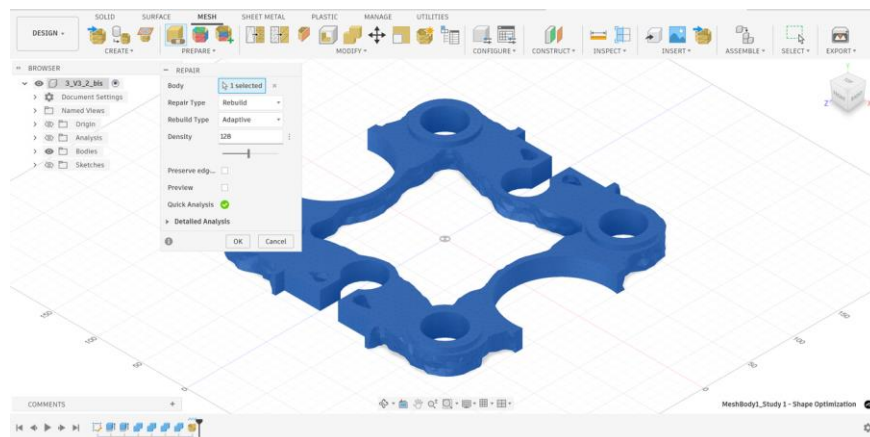


Figure 16. Mesh repair (Rebuild / Adaptive) to obtain a clean, oriented shell.

5.2 Remeshing and Facet-Count Control

With a clean shell available, the mesh density is tuned using the Remesh and Reduce commands (adaptive type, controlled by a proportion / density value). Adaptive remeshing uses few triangles on flat areas and more on the curved regions, which is exactly what an organic shape needs. The facet count reported in the Properties panel changes accordingly as the surface is refined, while the mass stays essentially constant – confirming that remeshing changes only the surface discretization, not the amount of material:

Reconstruction stage	Facet count	Vertex count	Mass (g)
After repair (coarse)	9,506	4,737	4,563.8
Intermediate	13,204	6,586	4,568.6
Refined	52,820	26,394	4,567.3
Final (smooth)	102,546	51,257	4,570.5

The facet count therefore rises from about 9,500 to roughly 102,500 triangles as the surface is progressively refined, giving a smooth, well-defined skin for the conversion, while the mass remains at about 4.57 kg throughout (consistent with the 4.582 kg of the optimized result). The density (about 0.008 g/mm³, i.e. 8 g/cm³) and the steel material are preserved at every stage.

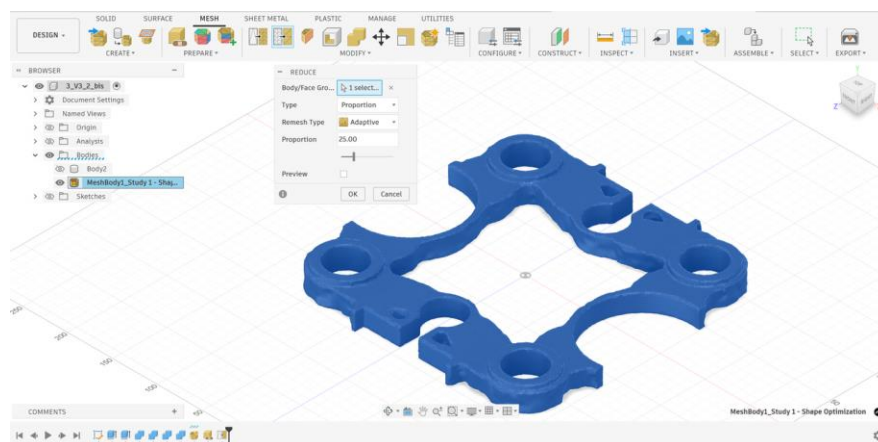


Figure 17. Reduce / remesh operations (adaptive) used to control the facet count.

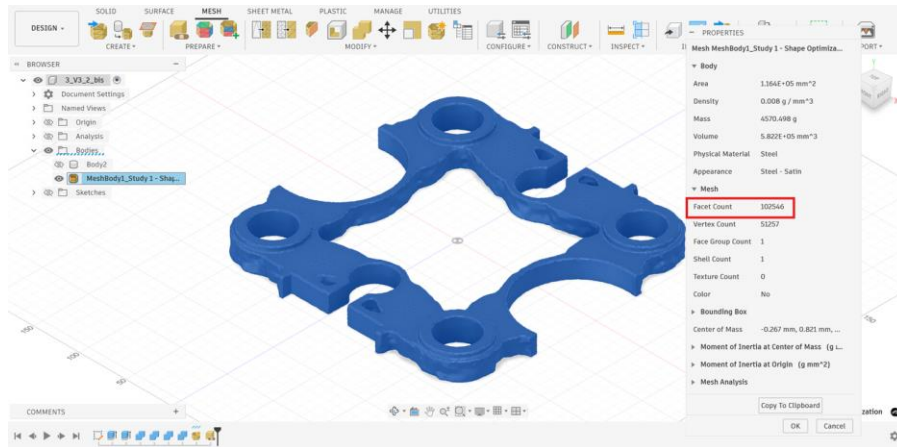


Figure 18. Mesh Properties showing the refined facet count (102,546).

5.3 Conversion to a Solid Body

The refined mesh is finally turned into a solid using the Convert Mesh command (parametric operation). Fusion offers different conversion methods: a Faceted method, which transfers every triangle directly into a flat face and produces a heavy, angular prismatic solid, and an Organic method, which fits smooth free-form surfaces over the mesh. The faceted route was tried but is poorly suited to this geometry; the Organic method was chosen instead, since it reconstructs the curved, branch-like shape faithfully and yields a clean smooth solid. It is more complex for the software to compute, but it is the only option that gives a result usable as a real engineering body.

Convert Mesh method	Outcome
Faceted	Direct triangle-to-face, angular prismatic solid – not used
Organic	Smooth free-form solid following the optimized shape – used for the final body

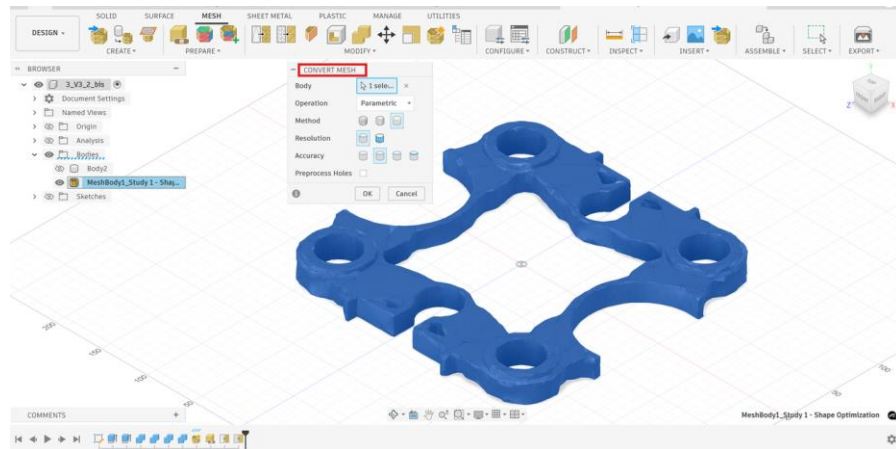


Figure 19. Convert Mesh dialog (Organic method selected).

5.4 Final Reconstructed Body

The output of the conversion is the final solid body: a smooth, organic version of the optimized skeleton, with the four functional bores preserved and connected by the curved ribs produced by the optimization. This geometry could not realistically have been produced with conventional CAD or machining – it is the kind of free-form shape that justifies the additive-manufacturing route – and it is the body taken forward to the static structural validation.



Figure 20. Final reconstructed organic solid body.

5.5 Recreation of the Functional Bores

The organic conversion produces a smooth solid, but it also rounds and slightly distorts the cylindrical bores, because the converter fits free-form surfaces over the whole mesh. Since the four

bores are the functional features of the part (they define how it is mounted and loaded), they are re-cut cleanly through the body with the Extrude command in Cut mode. Four circular profiles are selected and extruded symmetrically through the solid (Operation: Cut), which removes the rounded material and restores true, sharp cylindrical holes. Symmetric cuts are used so the holes are clean from both faces of the part.

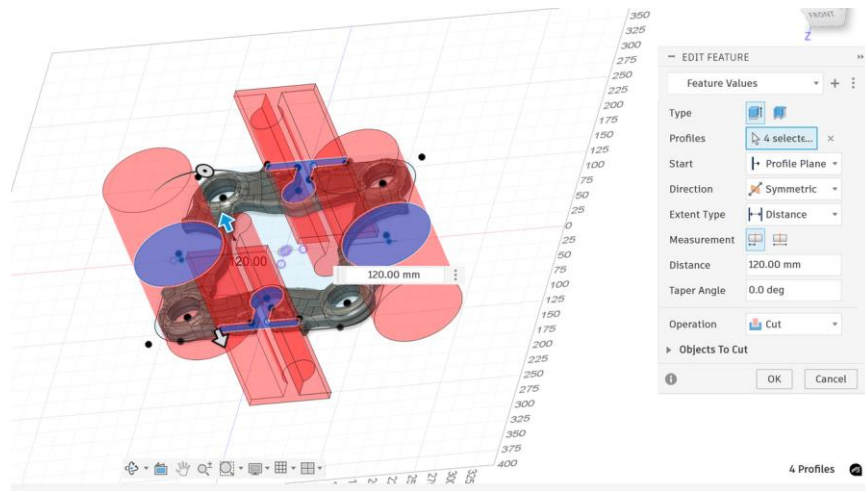


Figure 21. Recreating the four bores with a symmetric cut-extrude through the organic solid.

5.6 Mass Comparison (Original vs Optimized)

With the final solid complete, the achieved weight saving can be quantified directly by comparing the mass of the original plate (kept in the file as Body2) with the mass of the optimized body, both evaluated in the same steel (density 0.008 g/mm³):

Configuration	Mass (g)	Mass (kg)	Volume (mm ³)
Original plate (Body2)	8 935.5	8.94	1.138×10^6
Optimized final body	4 506.5	4.51	5.741×10^5
Reduction	$\approx 4\,429$	≈ 4.43	$\approx 49.6 \%$

The optimized component therefore weighs about 4.51 kg against 8.94 kg for the original plate – a mass reduction of roughly 49.6%, essentially the 50% target that was set at the start. The retained-mass figure agrees with the 51.22% mass ratio reported by the optimizer and with the body mass tracked during reconstruction (≈ 4.5 kg), confirming the whole reconstruction was mass-consistent. It is worth noting that these values refer to the steel design study; the part is later 3D-

printed in polymer, so the physical printed object will be far lighter (this is discussed in the additive-manufacturing section).

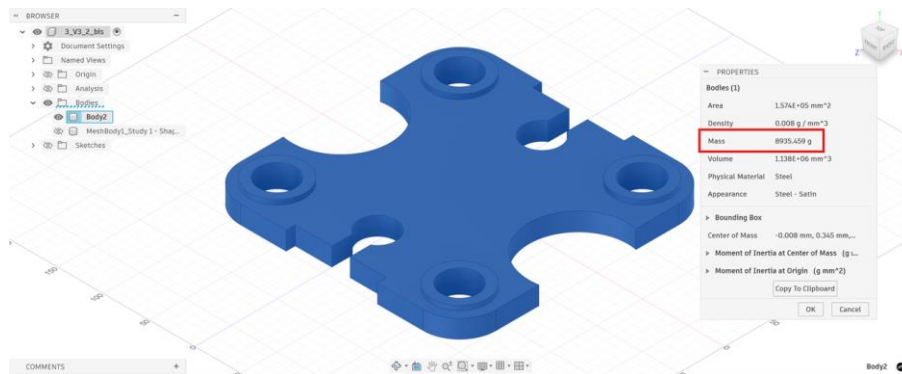


Figure 22. Mass of the original plate (Body2): 8 935 g.



Figure 23. Mass of the final optimized body: 4 507 g.

6. Static Structural Validation

Shape optimization is only a design tool: it produces an efficient shape under the criteria it was given, but it does not by itself prove that the resulting part is safe. To confirm that the lightened component can still carry the design loads, a separate Static Stress study (Study 2) is set up on the reconstructed solid, using the same loading scenario as the optimization. This study computes the real stresses and displacements in the optimized geometry.

6.1 Study Setup

The static study reuses the boundary conditions of the optimization so that the comparison is meaningful:

Material – the same steel as before, with the safety factor referenced to yield strength.

Constraints – a Fixed constraint is applied to the four bore faces, locking all three translational degrees of freedom (U_x , U_y , U_z). These faces are the anchoring points of the part.

Loads – the same four forces used in the optimization are applied to the four bores, acting in the plane of the part along the diagonal directions; gravity is again left disabled, since the self-weight is negligible next to the applied forces.

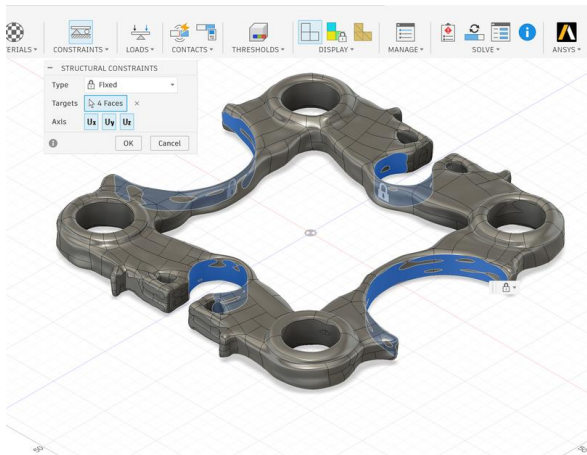


Figure 24. Static study constraints: the four bore faces fully fixed (U_x , U_y , U_z).

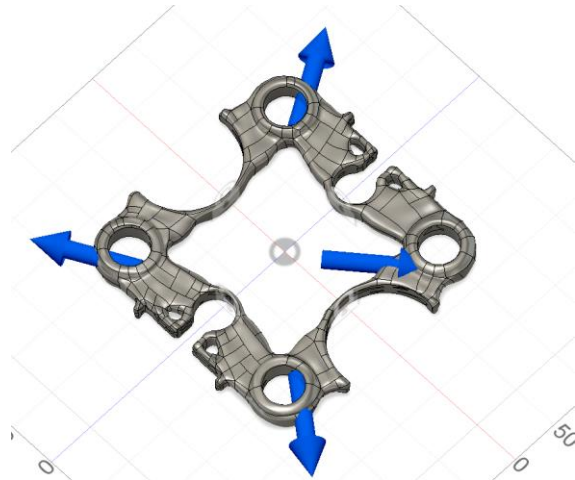


Figure 25. Static study loads: the four in-plane diagonal forces on the bores.

6.2 Degrees-of-Freedom Check

As in the optimization, the model is checked with the Degrees of Freedom display before solving. The optimized body appears entirely green (fully fixed), confirming that the constraints fully restrain the part with no rigid-body motion. The problem is therefore well posed and ready to solve.

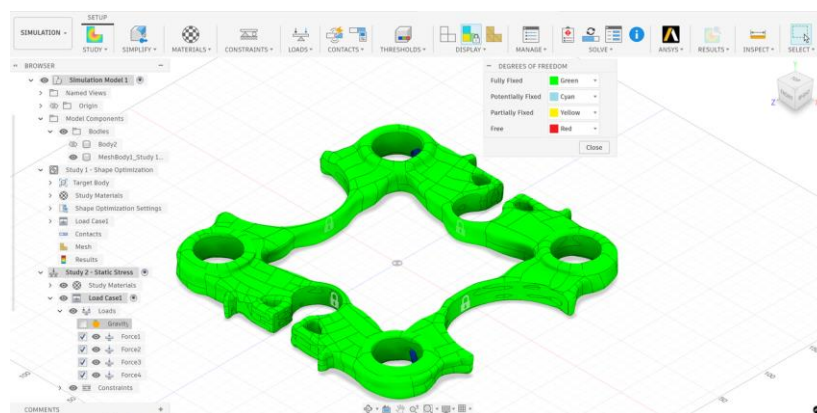


Figure 26. Degrees-of-freedom check on the optimized body: fully fixed (green).

6.3 Solving

The static stress study is submitted to the cloud solver (the structural solve runs through Fusion's cloud simulation service). The Job Status panel shows Study 2 – Static Stress running on Simulation Model 1, while the earlier shape-optimization study is listed as complete. Once the solve finishes, the results (von Mises stress, displacement and safety factor) are fetched back into Fusion for evaluation, which is the subject of the next section.

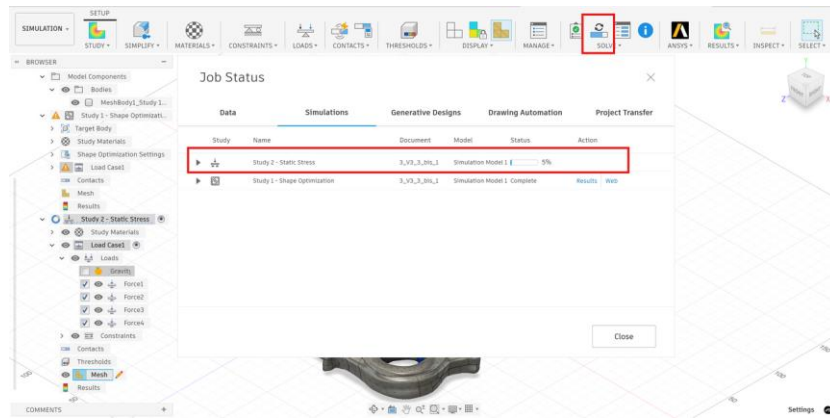


Figure 27. Cloud solve of the static stress study (Job Status).

6.4 Static Stress Results

Once the cloud solve completes, the results are reviewed under the Results branch of Study 2 (von Mises stress, displacement, strain and safety factor). For all plots, the legend is essential to read the values, since the colours are only meaningful against their numeric scale.

Result	Maximum value	Interpretation
Von Mises stress	6.587 MPa	Far below the steel yield strength
Total displacement	7.082×10^{-4} mm	Negligible deflection (sub-micron)
Equivalent strain	3.706×10^{-5}	Very small elastic strain
Safety factor	Above target everywhere (≥ 4)	Large margin against yielding

The von Mises stress reaches only about 6.6 MPa at its worst point, with most of the part deep in the blue (low-stress) range; this is a very small stress for steel. The total displacement is of the

order of 7×10^{-4} mm, i.e. well under one micron, so the part is effectively rigid under the load case. The safety-factor plot is the clearest summary: the whole body is in the “Above Target” (blue) band of the threshold scale (target band 2–4, display capped at 8), meaning the safety factor lies comfortably above the target everywhere, with no region falling below it. Taken together, the results confirm that the lightened, optimized geometry carries the design loads with a large margin – **the optimization removed roughly half the mass without compromising the structural integrity of the part.**

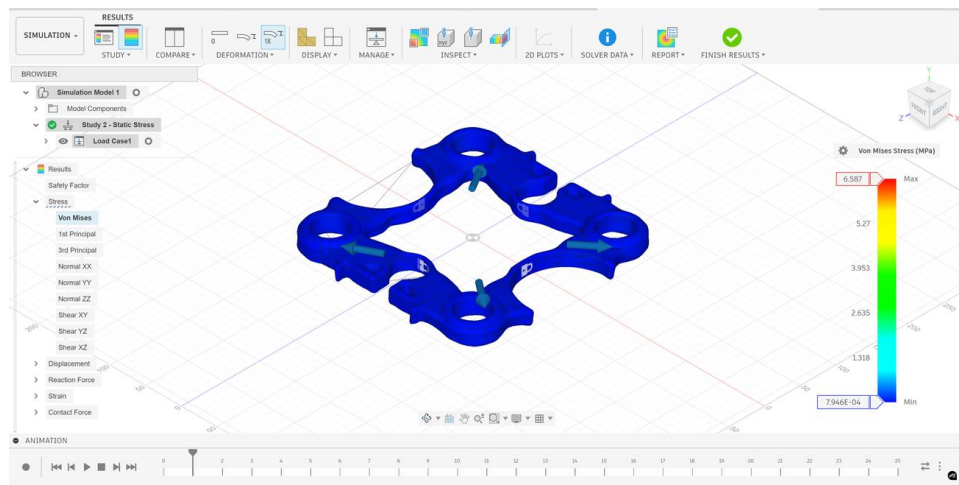


Figure 28. Von Mises stress (max 6.587 MPa); the part is almost entirely in the low-stress range.

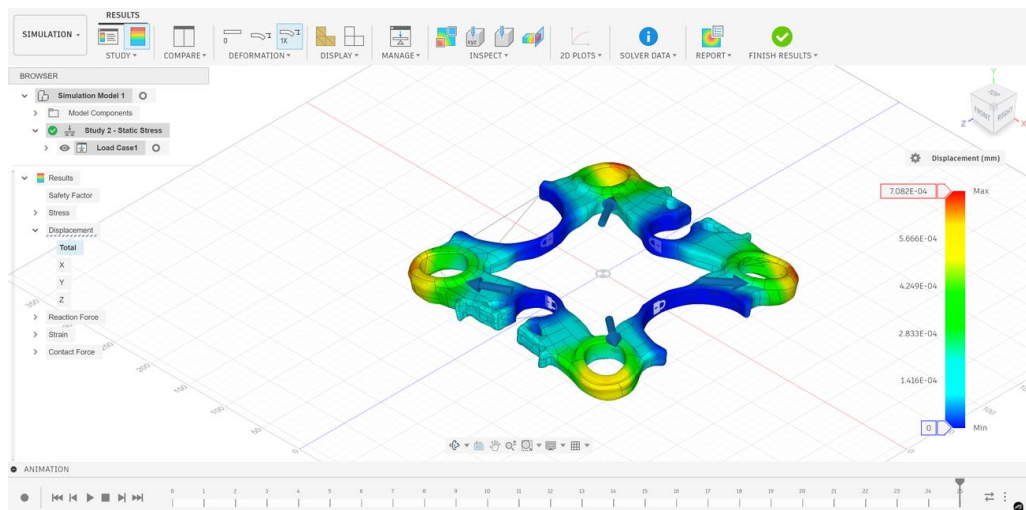


Figure 29. Total displacement (max 7.08×10^{-4} mm).

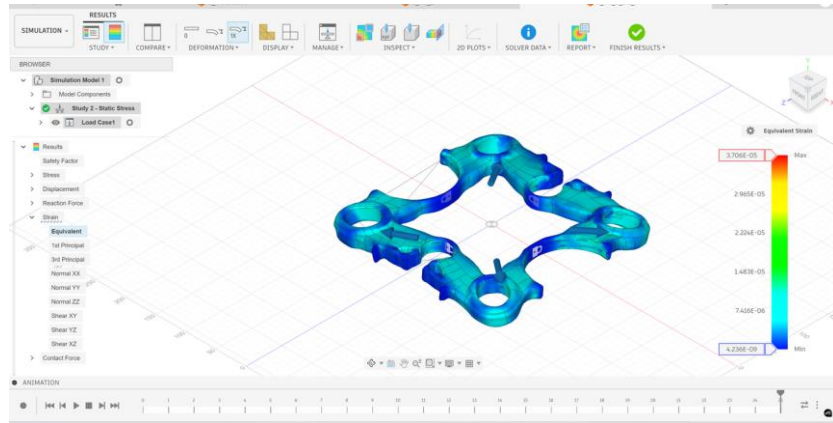


Figure 30. Equivalent strain ($\max 3.71 \times 10^{-5}$).

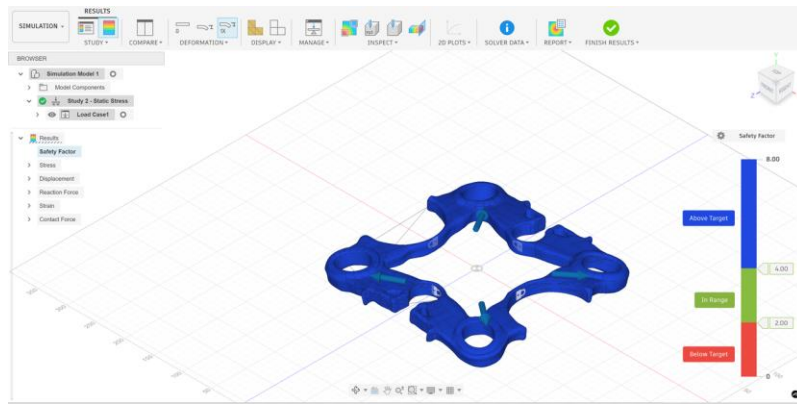


Figure 31. Safety factor: the whole part lies above the target band (large margin).

7. Additive Manufacturing Preparation

With the geometry validated, the part is prepared for 3D printing. As explained at the start, the organic optimized shape is essentially impossible to make with conventional processes, so additive manufacturing (FFF / FDM) is the natural route. The preparation is done first in the Manufacture workspace of Fusion (Additive) and then verified in a dedicated slicer, **PrusaSlicer**.

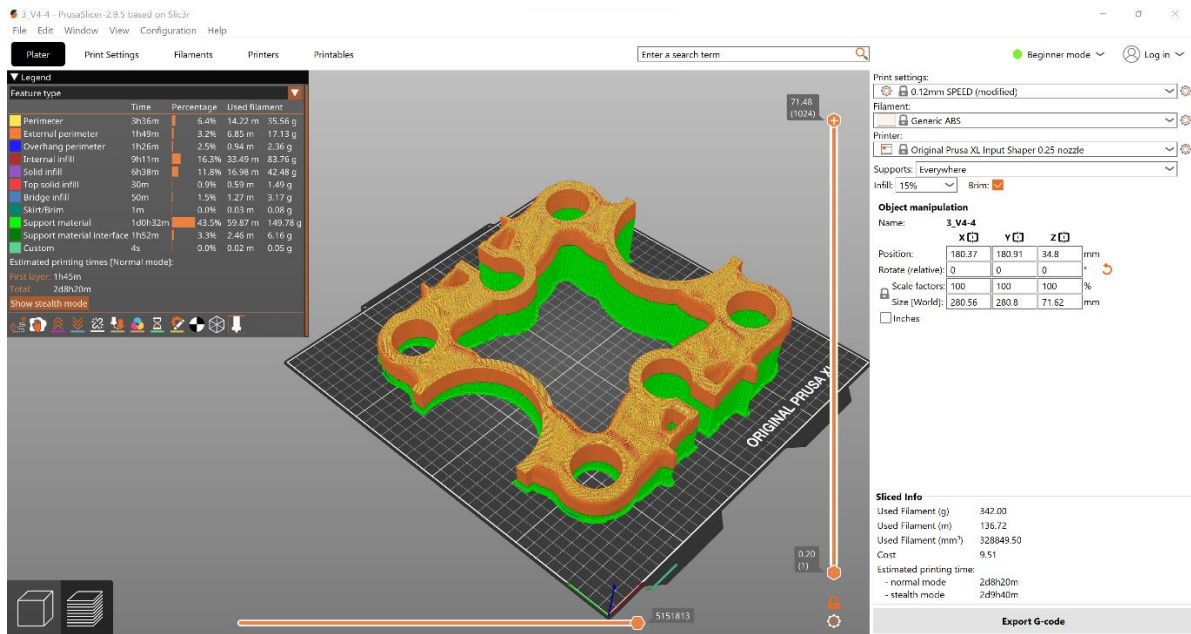


Figure 32. PrusaSlicer.

7.1 Additive Setup in Fusion

In the Manufacture workspace an additive Setup is created on the optimized body. The main choices are:

Item	Setting
Operation type	Additive (with automatic arrangement on the plate)
Machine	Prusa XL – FFF, single tool, build volume $360 \times 360 \times 360$ mm
Print setting	ABS (Direct Drive) – FFF, 0.15 mm layer height, 0.4 mm nozzle
Supports	Solid Volume Support, overhang angle 40°

The machine is selected from the library (**Prusa XL**) and the material print setting (ABS, Direct Drive) is applied. Because the optimized part has many curved, branch-like overhangs, supports are required: a Solid Volume Support is generated with a 40° overhang angle so that all faces steeper than this threshold are supported. An additive toolpath is then created and checked with the Simulate Additive Toolpath tool, which previews the part layer by layer together with its supports before any post-processing.

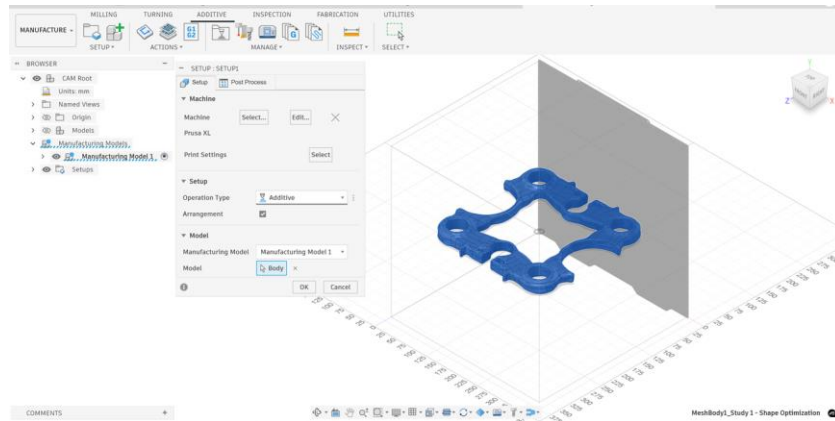


Figure 33. Additive setup (operation type Additive, model = optimized body).

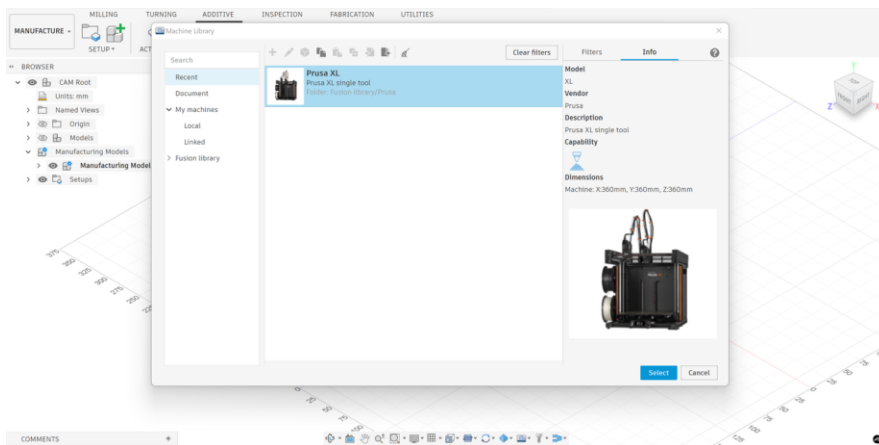


Figure 34. Machine selection: Prusa XL (FFF, 360 × 360 × 360 mm).

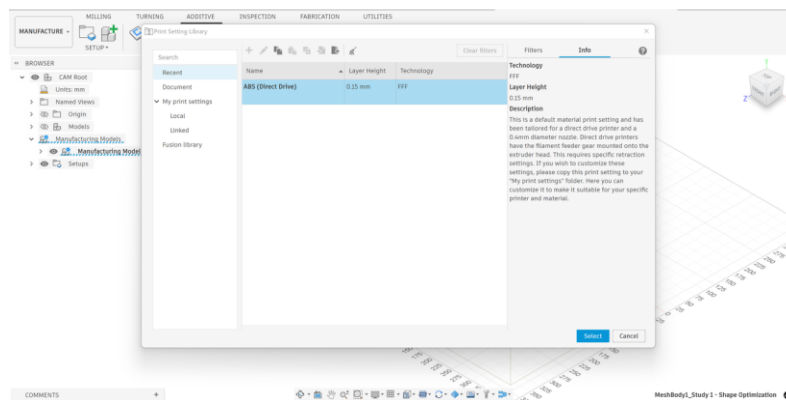


Figure 35. Print setting: ABS (Direct Drive), 0.15 mm layer height.

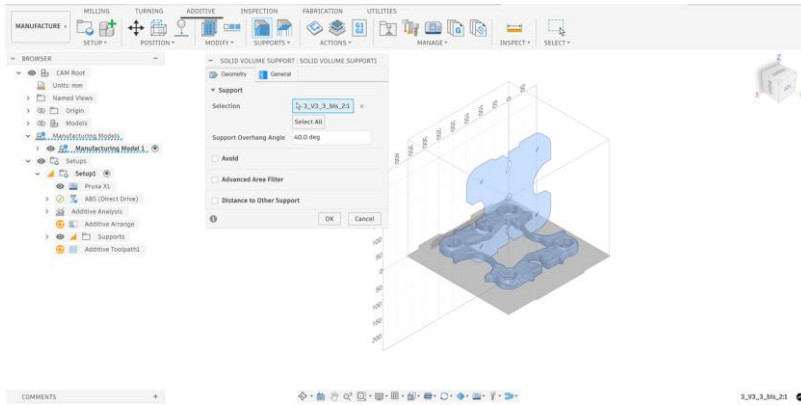


Figure 36. Solid Volume Support generation (overhang angle 40°).

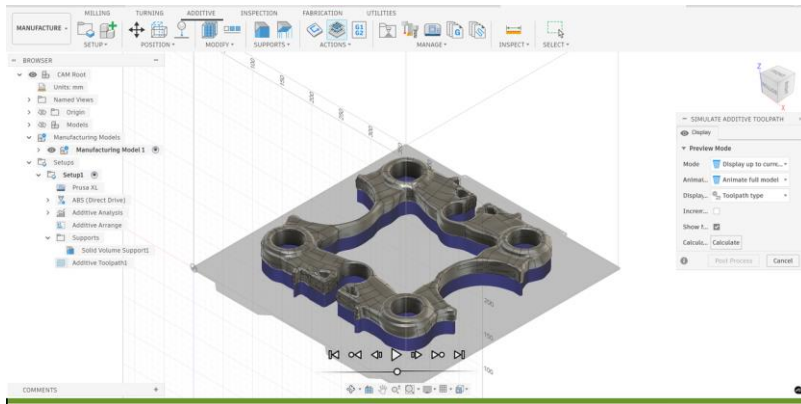


Figure 37. Additive toolpath simulation with supports.

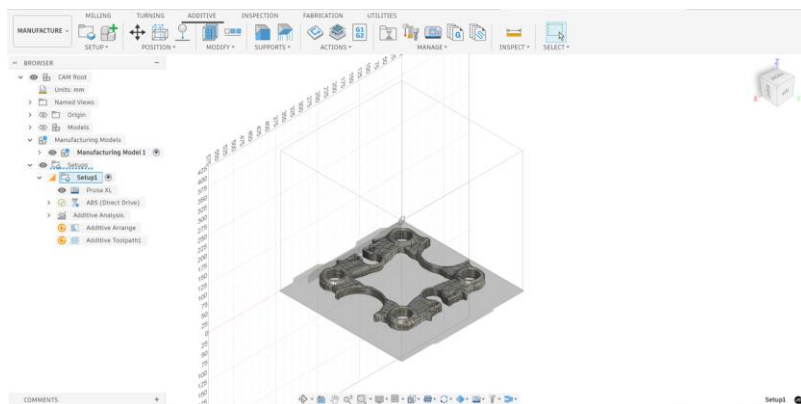


Figure 38. The optimized part arranged on the build plate.

7.2 Slicing and Verification in PrusaSlicer

Fusion's additive environment prepares the part and a first toolpath, but it works on an idealized model. To see the real, practical print outcome, the part is exported and opened in PrusaSlicer, the dedicated slicer for the printer. A dedicated slicer is tuned to the specific machine and filament and reports the true build reality – the actual support volume, material mass, print time and cost – which the CAD additive preview does not expose. This is why the output checked in PrusaSlicer is a more faithful picture of manufacturability than the Fusion preview. The slicing was set up as follows:

Parameter	Value
Print profile	0.12 mm SPEED (modified)
Filament	Generic ABS
Printer	Original Prusa XL (Input Shaper), 0.25 mm nozzle
Supports / infill / brim	Everywhere / 15 % / yes
Part size	280.56 × 280.80 × 71.62 mm
Material used	342.00 g (136.72 m, 328 850 mm ³)
Estimated cost	9.51
Print time	2 d 8 h 20 m (normal) / 2 d 9 h 40 m (stealth)

The slicer also breaks the job down by feature, which reveals the most important manufacturing lesson of the whole project:

Feature	Time	Share	Filament
Support material	1 d 0 h 32 m	43.5 %	149.78 g
Internal infill	9 h 11 m	16.3 %	83.76 g

Feature	Time	Share	Filament
Solid infill	6 h 38 m	11.8 %	42.48 g
Perimeter	3 h 36 m	6.4 %	35.56 g
External perimeter	1 h 49 m	3.2 %	17.13 g
Support interface	1 h 52 m	3.3 %	6.16 g

Support material alone accounts for 43.5% of the print – about 150 g of the 342 g total and a full day of the print time. This is the practical price of the organic optimized shape: a geometry that is excellent structurally (light and stiff) is expensive to print, because its overhangs and free-form ribs need a large amount of support. It is exactly the kind of trade-off that only becomes visible in the slicer.

The estimated print time of about 2 days and 8 hours is high, but it is realistic for this setup and is driven by the chosen parameters rather than by any error. Three factors dominate it. First, the print uses very fine settings – a 0.25 mm nozzle with 0.12 mm layers – so the ≈ 72 mm height is built in roughly 600 thin layers of narrow extrusion lines; this gives good detail but is inherently slow. Second, the supports account for 43.5% of the job (almost a full day on their own), because the organic optimized shape has many overhangs that must be scaffolded. Third, the part has a large footprint ($\approx 280 \times 280$ mm), and print time scales with volume and area. Multi-day prints are normal for large or detailed FDM parts, so the figure is not anomalous. If the part had to be produced in practice, the time could be reduced substantially – for example by using a larger nozzle (0.4 mm) and thicker layers (0.2 mm), by re-orienting or splitting the part to cut the amount of support, and by reducing the infill where strength allows. For this demonstrator, however, the result is reported as-is, since it usefully exposes the design-for-additive-manufacturing trade-off: the optimization makes the part light and stiff, but the resulting free-form geometry is expensive to print, and only the slicer makes that cost visible.

It is also worth separating the two mass figures. The optimization and validation were carried out in steel, where the body weighs about 4.5 kg; the physical part, however, is 3D-printed in ABS and uses only about 342 g of filament (supports included). The two values describe different things – the steel design study versus the polymer demonstrator.

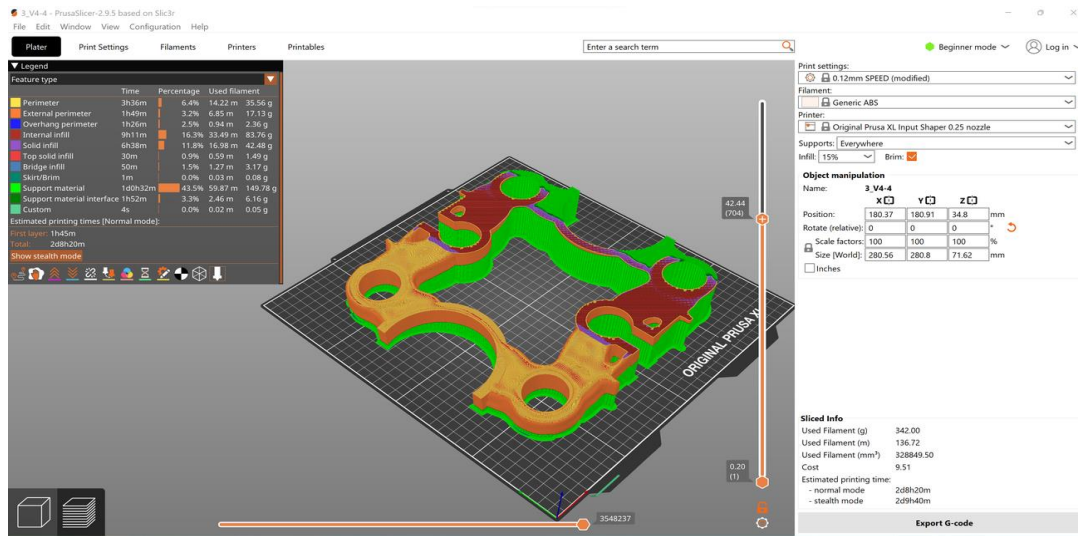


Figure 39 Sliced part in PrusaSlicer: feature breakdown, material (342 g), time (≈ 2 d 8 h) and cost; support material is 43.5%.

8. Conclusions

This project followed a complete design-for-additive-manufacturing workflow on a steel plate component, from lightweighting through to a printable part:

- Shape optimization reduced the mass to the 50% target: the optimized body weighs about 4.51 kg against 8.94 kg for the original plate, a reduction of roughly 49.6% (mass ratio 51.22%).
- Reconstruction turned the coarse optimization mesh into a clean organic solid (mesh repair, adaptive remeshing, organic conversion and re-cutting of the bores), keeping the mass consistent throughout.
- Static structural validation confirmed the lightened part is safe under the design load case: a maximum von Mises stress of only 6.6 MPa, sub-micron displacement, and a safety factor everywhere above the target band.
- Additive preparation and slicing produced a printable job for the Prusa XL in ABS, and – importantly – quantified the real cost of the organic shape: a print time of about 2 d 8 h and 342 g of filament, of which 43.5% is support material.

The main engineering takeaway is twofold. First, shape optimization plus additive manufacturing makes it possible to obtain a much lighter, structurally efficient part whose free-form geometry could not be produced by conventional means. Second, the optimized shape is not “free” to

manufacture: a dedicated slicer such as **PrusaSlicer** is needed to reveal the true build reality – support volume, material, time and cost – and in this case it showed that supports dominate the job. The project therefore illustrates the full loop of modern part design: optimize the geometry, validate it structurally, and then confront it with the real constraints of the manufacturing process.